

***Los Angeles Basin Smog of 1943: A History of Air Pollution in an Arid Industrial Valley  
Region***

By Fiona Keady

It was July 26, 1943, when the city of Los Angeles was covered in a sheet of smog. Those who were affected believed that this smog was caused by an attack by opposing countries in World War II, but soon discovered that the smog had been sourced by various sources within their own city: “smoke-belching locomotives and diesel trucks, burning rubbish in back yard incinerators and city dumps and the combustion of scrap lumber in sawmills” (South Coast AQMD, 1997). It was not truly a new issue in the incident of 1943. Since 1903, industrial plants have been filling the basin with emissions that looked like heavy clouds of smoke. The conditions of the Los Angeles Basin region are part of the reason why this event was so impactful, but also the public policy at the time in accordance with air quality standards has not helped either.

The primary emission sources in this issue are known as precursors to Ozone emissions: Nitrogen Oxides (NOx) and Volatile Organic Carbons (VOCs). NOx is usually formed from atmospheric chemical reactions driven by sun rays. These particulates are the major cause of pollution events in the Los Angeles Basin. Reactive Hydrocarbons (RHCs) are a unique pollutant in the case of the basin, as they are emitted in the form of aerosols into the atmosphere. There are two types of aerosols: Primary and secondary. Primary organic aerosols (POAs) are typically sourced from combustion processes such as car exhaust, cooking, and wildfires. Secondary organic aerosols (SOAs) are formed indirectly due to reactions from other emitted pollutants with sunlight (Kiest, 2023). All in all, it seems that sunlight has a major effect on the pollution of the Los Angeles Basin’s air. On June 27, 1974, topographical bias data was taken (Figure 1) to measure the NOx emissions in the San Bernardino area to measure exactly the effects these emissions were having on different parts of the region.

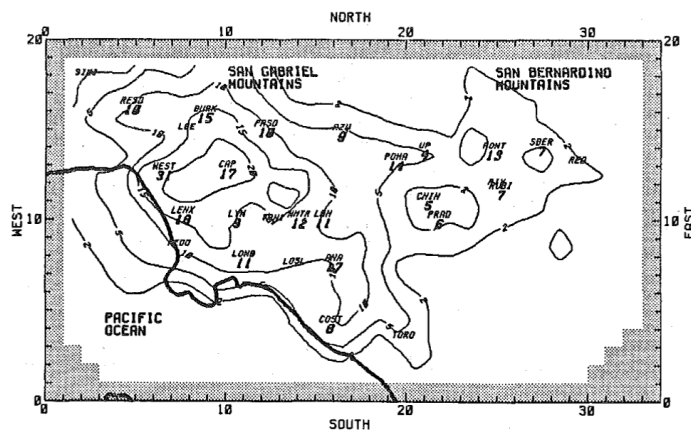


Figure 1. Topographical dispersion map of NOx emissions (in pphm) (Teschke et al., 1984)

The meteorological and geographical conditions of the LA Basin also have a major effect on its emissions. As mentioned before, heat and sunlight have a major impact on these pollution conditions. Los Angeles is a very dry and hot area of the United States: research indicates that these meteorological conditions will cause higher particulate concentration levels in the air. There was a study done from 2016-2018 to measure the amounts of PM2.5 levels at different

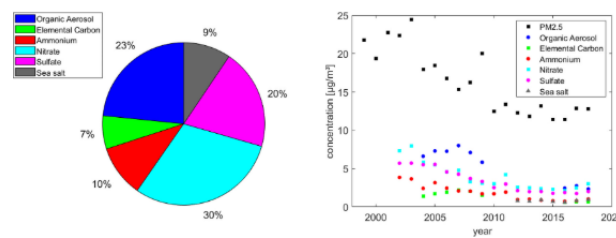
temperatures in the Los Angeles basin. The findings of this study found that at 68 degrees Fahrenheit, there were nearly zero PM<sub>2.5</sub> concentration violations to air quality standards; but in 70-80% of the days where the temperature exceeded 100 degrees Fahrenheit, the PM<sub>2.5</sub> levels exceeded National Ambient Air Quality Standards (Kiest, 2023). The South Coast AQMD recognized that the sea breeze also has effects on the dispersion of these PM<sub>2.5</sub> pollutants. Particularly from the coast inwards, which can also be observed from Figure 1. The figure presents that the concentrations of NO<sub>x</sub> on the coast are higher, but the predominant wind direction of northwest wind is also affected with high concentrations of NO<sub>x</sub> emissions.

Experts have used modeling methods and legislative methods to estimate and reduce pollutant emissions. Since the early 1970s, early applications used the Systems Application Airshed Model to categorize the properties of reactive hydrocarbons. How this model works is that it is simulated through the atmospheric diffusion equation:  $\frac{\partial C_l}{\partial t} + U_i \frac{\partial C_l}{\partial x_i} = \frac{\partial}{\partial x_i} K_i \left( \frac{\partial C_l}{\partial x_i} \right) + S_l + R_l$  where  $C_l$  is the approximate value of the instantaneous concentration of pollutant “l” at a point,  $U_i$  is the mean wind velocity in that area at that time,  $K_i$  represents the eddy diffusivity components,  $S_l$  is the rate of emission of pollutant “l” from elevated sources or stacks, and  $R_l$  stands for the contribution of the mean concentration to the net reaction rate of the pollutant (Tesche et al., 1984). Using this equation, the city of Los Angeles has kept inventories of pollutant reactions occurring in the air since the year 1974. They have used this data as a performance evaluation for control mechanisms. This has assisted them in finding the source of their problem and finding solutions through legislative action.

The history of the city of Los Angeles combating air pollution through their regulations begins in the late 1950s. In 1947, the Los Angeles County Board of Supervisors established the first air pollution control program in the nation. In 1959, the California Motor Vehicle Control board was created to combat the issue of not being able to regulate motor vehicle pollution. This organization was given the authority by the government to test pollutant emissions and certify control devices. In 1977, the South Coast Air Quality Management District (AQMD) was created as a combination of four other air quality management boards within the city. The AQMD has been continuously making efforts to manage pollution control over the years. In 1978, they added cumbersome sleeves on gas pump nozzles to prevent hydrocarbon gases from escaping. In the late 1980s, they adopted measures to control pollutants that were seen as a threat to the stratospheric ozone layer such as chromium and asbestos. In 1993, they created RECLAIM, the Regional Clean Air Incentives Market which provides regulations for each piece of equipment that emits pollutants into the air and continues to reduce the emissions limit each year (South Coast AQMD, 1997). As of 2024, the limit for nitrogen oxides for unregulated emission sources such as drying ovens and roasters is 30 ppmv.

Data presents that improvements to combating these emissions have been successful. From 1977 to 1996, stage 1 ozone episodes have been reduced from 121 to just 7, and have completely disappeared by 2024 (South Coast AQMD, 1997). Yet there is another underlying problem with

pollutants in this area. From 1999 to 2012, studies have discovered that particulate matter with 2.5um diameter (PM2.5) emissions have been halved. Since 2012, these emission levels have remained almost unchanged (Figure 2). These PM2.5 emissions are compared to the emissions of multiple pollutants: organic aerosols, carbon, ammonium, nitrate, sulfate, and sea salt. Many of these are not a concern due to natural circumstances, but the main issue air quality experts are focusing on currently are aerosols, NOx, and PM2.5, which are the highest levels according to the data. Though these levels are being reduced over time, the PM2.5 levels are still significantly higher than they should be in this area.



Aerosol concentrations in LA have declined over time, leveling out around 2012. Credit: Nussbaumer and Cohen, Environ. Sci. Technol. 2021

*Figure 2. 1996 – 2020 Emission Concentrations in Los Angeles (Kiest, 2023)*

Since the major smog events occurring between the 1950s throughout the 1970s, the city of Los Angeles has worked relentlessly over the years to mitigate the effects of pollutant emissions in this area. Driven by both natural and man-made factors, these pollution events have been the subject of extensive research and regulatory efforts for almost a century. Although substantial reductions in ozone episodes and NOx emissions have been achieved, PM2.5 remains a significant concern, with concentrations consistently exceeding established air quality standards as well as remaining constant over the past decade. Despite the implementation of advanced modeling techniques and stringent legislative measures, the interaction of the region's topography, semi-arid climate, and extensive industrial activities continues to impede the efforts to mitigate air pollution in the Los Angeles basin. Moving forward, further advancements in emission control technologies and public policy will be essential to effectively address the persistent challenges posed by fine particulate matter and the ozone to safeguard public health in this region and prevent major smog events in this region from happening again.

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